



Click to Chat
my AI Twin

Vishal Khan

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MSc student in **AI & Computer Science at Northeastern University London**, with a prior MSc in **Data Science from LJMU**. Around **2 years** as a Data Engineer at Teleperformance plus **2+ years** translating commercial requirements into analytics and **ML forecasting models**. Comfortable across the full analytics chain, framing problems, building models in **Python**, and shipping end-to-end applications. MSc thesis compared **Reinforcement Learning** for dynamic pricing in e-commerce.

Tools & Skills

Programming	Python, SQL, R, HTML, CSS, JavaScript
Machine Learning	Reinforcement Learning, Deep Learning, Classification, Regression, Time-Series Forecasting, NLP, Feature Engineering, Model Evaluation
Generative AI	Fine-Tuning LLMs, RAG, Prompt Engineering, LangChain, Agentic AI, MCP
Libraries	PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Statsmodels, Matplotlib, Seaborn, Plotly
Data & Cloud	ETL Pipelines, Databricks, Azure, AWS, MySQL, SQL Server, MongoDB, Neo4j
Viz & BI	Power BI, Tableau, Excel
Tools	Git/GitHub, VS Code, Jupyter, Google Colab, Streamlit, Vercel, Railway

Experience

Teleperformance Global Business Pvt. Ltd. *June 2024–Jan 2026*
Data Engineer

- **Built and maintained Databricks pipelines** processing **millions of telecom records** (CDRs, billing, agent performance), feeding a data engineering team of **50+ across multiple regions**.
- **Migrated** on-prem **SQL Server** schemas to **Databricks**, improving query performance by **~20%** and cutting infrastructure costs by **~20%**.
- **Partnered** with analytics and reporting stakeholders to translate ambiguous business questions into pipeline requirements, surfacing data they could actually use for decisions.

Avant Garde Clean Room & Engg. Solutions Pvt. Ltd. *Aug 2021–Dec 2023*
Executive

- **Built ML sales forecasting models** reaching **80–90% accuracy** on monthly revenue projections, used **daily** by the business for commercial planning.
- **Translated** commercial questions into analytical workflows in **Python and SQL**, turning ad-hoc requests into recurring reports the team relied on.
- **Rebuilt** revenue and collections reporting in advanced **Excel (formulas, macros, VBA)**, cutting monthly reporting time from **~2 hour to ~30 minutes**.

Aryan Care Pvt. Ltd. *Jan 2021–Aug 2021*
E-commerce & Finance Executive

- Managed GST, ITR filings, and audits

PJRJ Associates *Jul 2018–Sep 2020*
Finance/Accounts/Taxation

- **2 years of finance and taxation operations experience**

Education

Northeastern University London, UK *Jan 2026–Jan 2027*
M. Sc. in AI and CS

Liverpool John Moores University, UK

Mar 2023–Mar 2024

M. Sc. in Data Science

IIIT Bangalore, India

Jan 2022–Jan 2023

Executive PG Program in Data Science

University of Delhi

Nov 2018–Nov 2021

Bachelor of Commerce

Projects

AI Digital Twin — Full-Stack RAG Assistant (Generative AI)

- **Objective:** Rebuild a vanilla-JS prototype into a production digital twin answering recruiter and technical-interview questions from grounded, cited evidence.
- **Approach:** **FastAPI** + **ChromaDB** RAG over 87 chunks (YAML, resume PDF, live **GitHub** ingestion) on **Azure OpenAI**; intent-boosted retrieval with conversation-aware follow-ups, **SSE** streaming, tool calling, **React 19** UI.
- **Result:** Shipped on **Railway/Vercel**; non-blocking content-hash ingestion holds health checks at ~20 ms. Adds rate limiting, daily spend cap, prompt-injection defences, **178 tests** in CI with **mypy**.

Image Classification (Deep Learning)

- **Objective:** Develop a Convolutional Neural Network model for image classification on the CIFAR-10 dataset
- **Approach:** Designed a **CNN in TensorFlow/Keras** with normalization, dropout, and L2 regularization to enhance accuracy and generalization.
- **Result:** Achieve a training accuracy of **72.12%** and a validation accuracy of **74.06%**

Telecom Churn Prediction (Machine Learning)

- **Objective:** Predict customer churn from **21 predictor variables** and identify the strongest retention levers.
- **Approach:** Ran **EDA, preprocessing, and feature engineering in Python/SQL**, then benchmarked classification models including **Logistic Regression**.
- **Result:** Achieved ~**80% accuracy** and identified **tenure** as the strongest churn predictor.

Research

MSc Thesis: Reinforcement Learning for Dynamic Pricing in E-commerce *LJMU, 2024*

- **Objective:** Test whether RL can outperform pricing rules in a simulated e-commerce environment.
- **Approach:** Built a custom **Gym-style environment** on real e-commerce data, then trained **DQN, A2C, and PPO** agents over multi-episode simulations.
- **Result:** **DQN** produced the most stable profit curve. Wrote up the comparative analysis as part of the dissertation.

Certifications

- Data Science Programming Bootcamp
- Machine/Deep Learning Certificate
- SQL — HackerRank
- AWS Coursework — Great Learning
- Data Toolkit

Data Consent: I consent to the processing of my personal data for recruitment purposes under UK GDPR.

External Links: Selected headings include hyperlinks to supporting projects and credentials.